Executive Summary

Proactive Infrastructure Disruption Detection & Automated Incident Response - C-suite two-page brief

Strategic Message

The white paper proposes a shift from reactive, threshold-based infrastructure monitoring to a proactive, governed reliability platform that detects precursor signals, enriches incidents with context, and automates response through ML, GenAI and agentic AI. The central claim is that many production incidents leave detectable telemetry patterns 15-60 minutes before user impact; the opportunity is to convert those signals into earlier action, lower disruption, and less operational noise.

70%	40%+	<5%	>75%
MTTD reduction by Month 12	**MTTR improvement target**	**false positive rate**	**proactive detection coverage**

Why This Matters To The Executive Team

- **Operational resilience:** earlier detection reduces outage duration and customer-visible disruption before escalation into P1 events.
- **Cost and capacity:** alert noise reduction, automated triage and self-healing lower toil and keep scarce SRE/NOC capacity focused on high-judgement work.
- **Risk governance:** model registry, explainability, change control, immutable audit logs, data privacy controls and human approval gates make the automation governable.
- **Platform leverage:** the architecture builds on open standards and existing ITSM workflows, reducing vendor lock-in and adoption friction.

Proposed Operating Model

Layer	Executive Relevance	Core Capability
Unified telemetry	Creates enterprise-wide visibility across fragmented platforms.	OpenTelemetry, Kafka, schema normalisation, feature store.
ML detection	Moves the organisation from after-the-fact alerts to predictive signals.	Five-model ensemble with entity, peer, contextual and population baselines.
Incident orchestration	Reduces manual triage and accelerates coordinated response.	P1-P4 classification, enriched payloads, ServiceNow/PagerDuty/Jira/OpsGenie integration.
GenAI and agents	Turns telemetry into plain-language decisions and automates low-risk response.	Triage, diagnostic, communications, remediation, learning and orchestration agents.

12-Month Delivery Roadmap

Months 1-3 Foundation	**Months 4-6** Shadow mode	**Months 7-9** Controlled automation	**Months 10-12** Full automation
Deploy collectors, stand up pipeline, integrate top data sources, establish baselines and dashboards.	Train ML models, run shadow detections, tune suppression rules and validate ITSM integration.	Automate P1/P2 workflows, deploy advanced models, introduce diagnostic/remediation agents with review.	Automate all severities within policy, activate learning loop, report precision/recall and governance evidence.

Executive Value Case

Value Lever	Expected Outcome	How It Is Achieved
Customer and revenue protection	Reduced user-facing disruption and fewer severe incidents.	Telemetry precursors, cross-layer correlation, proactive P3/P4 intervention.
Faster recovery	40%+ MTTR improvement target.	Auto-created incidents, root-cause hypotheses, diagnostics and pre-linked runbooks.
Lower operational toil	30% alert-volume reduction and better analyst experience.	False-positive suppression, duplicate detection, GenAI summaries and automated communications.
Continuous improvement	Models and runbooks improve from real incident outcomes.	Bidirectional ITSM feedback, analyst labels, retraining and Learning Agent updates.

Governance, Risk And Control Posture

- **Explainable and auditable ML:** all models are registered, versioned, evaluated, shadow-tested before promotion and supported by SHAP-style rationale for anomaly scores.
- **Human control over automation:** actions are tiered from safe autonomous actions to human-only actions; Tier 3 actions require approval and Tier 4 actions are never executed by agents.
- **Hard guardrails:** rate limits, blast-radius caps, rollback requirements, change-freeze locks, prompt-injection defence, factual validation and kill switches are enforced outside the LLM.
- **Compliance readiness:** pseudonymisation, retention policies, SOC 2-aligned controls, GDPR Art. 22 positioning, data residency options and immutable audit logs support regulated use.

Key Decisions For C-Suite Sponsorship

Decision	Why It Matters	Recommended Direction
Pilot scope	Determines speed to value and evidence quality.	Select a critical but bounded production domain with clear ownership and usable telemetry.
Automation appetite	Sets risk boundary for autonomous remediation.	Approve augment-first rollout: GenAI summaries first, Tier 1-2 actions later, Tier 3+ gated.
Governance ownership	Prevents AI operations from becoming an unmanaged shadow process.	Create ML/AI operations governance with Technology, Risk, Security and Service Management representation.
Data foundation investment	Model quality depends on telemetry, CMDB and dependency-map hygiene.	Fund observability maturity assessment, CMDB service model cleanup and ITSM integration readiness.

Recommended Next Steps

- Run a 2-4 week observability maturity assessment across telemetry, CMDB, service dependency mapping and ITSM integration readiness.
- Nominate a pilot domain, executive sponsor, service owner and risk approver for the first 90-day foundation phase.
- Define KPI baselines before implementation: MTTD, MTTR, false-positive rate, alert volume, P1/P2 rate, feedback rate and model accuracy.
- Stand up the governance forum and approve the initial automation policy, safety tiers, audit requirements and human override model.

Bottom line: The proposal is not simply a monitoring upgrade. It is a reliability operating model that combines observability, ML, ITSM integration and governed AI automation to detect disruption earlier, resolve incidents faster and preserve human judgement for high-risk decisions.